INSPIRE - 100

An Emergency Ventilator Device

- *Unmatched Affordability*
- *Unmatched Remote Monitoring via WiFi*
- *Unmatched Ease-of-use*
- *No need for compressed Air Pipeline*
- *Connect to Oxygen Cylinder or Concentrator*
- *Full range of Mainstream Respiration Parameters*
- *Breath Synchronization for Patient Comfort*
- *Field Upgradeable*
- *Rugged and Robust*

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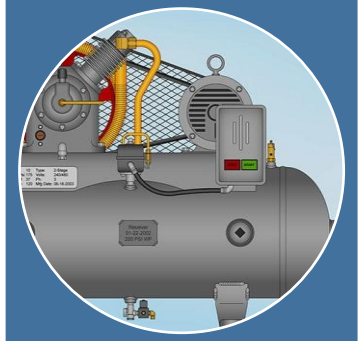
Setting the Context

The Problem Statement

The Motivation

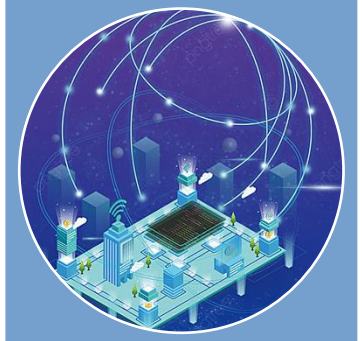
Evolution of Ventilators



Iron Lung Age

Pneumatic Age

µController Age



Smart "E"-Age

Most-used Ventilation Modes have not changed

- Volume and Pressure Control
- Control BPM, I/E, VT and PS
- Monitor pressures and flow
- Safety Alarm systems

Diminishing Returns from what has evolved …

- Exotic Ventilation modes
- Multitude of Sensors
- Touch-screen Displays

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The Problem
Scarcity & Affordability

India has amongst the lowest per capita ICU beds in the World*

- 1.46 beds / 1000 people*
- 3.65 ICU beds / Lakh people*
- Only half of ICU beds are equipped with Ventilators
- A meagre 1.8 Ventilators for one lakh people*

ICU ventilators are expensive equipment

- Unaffordable in remote clinics

Ventilator Ambulances are

- Non-existent except in few urban centers,

** As per April 2020 Study by Center for Disease Dynamics, Economics & Policy at Princeton University, USA*

Respiration Assist Devices

Categories – Usage and Pricing

Features

Less than Rs 50,000

CPAP

Less than Rs 1 Lakh

BiPAP

AFFORDABLE feature set for PHC, Small Clinic, and Ambulance Use?

Big Hole

Rs 10 Lakhs ++

ICU Ventilator

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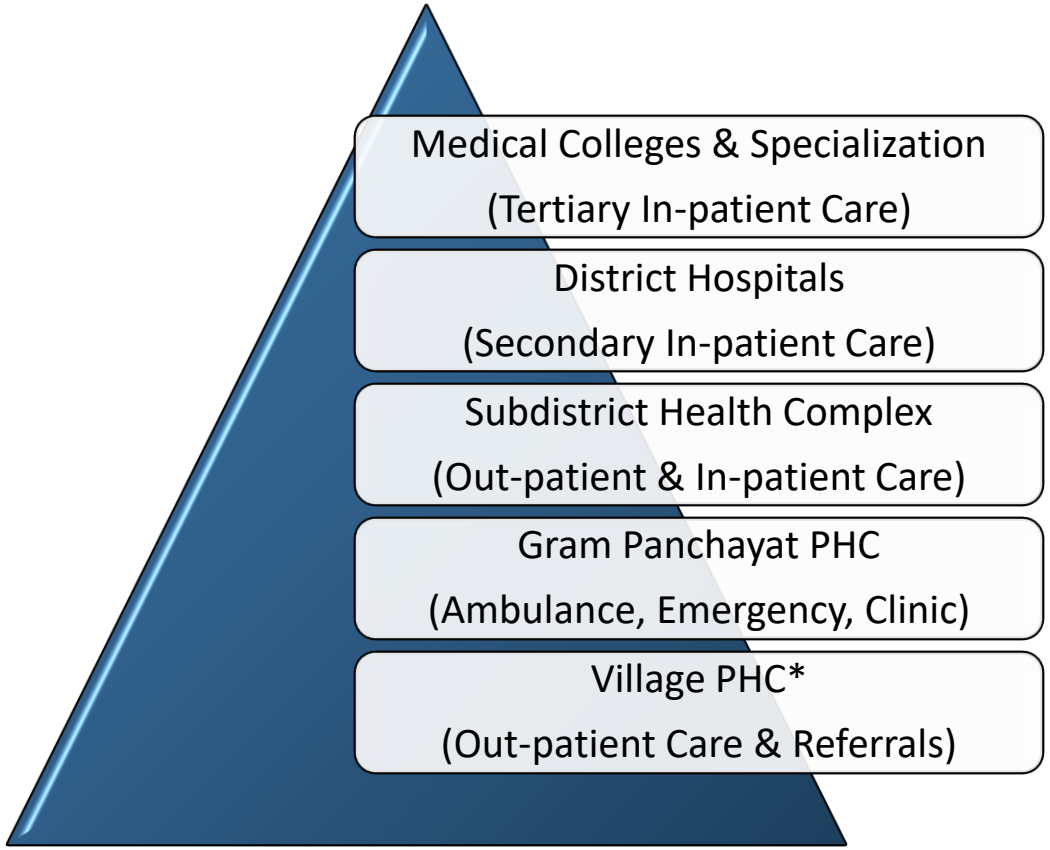
The Motivation



As of 2021, there were **1,56,231 Primary Health Centers (PHCs) in India with minimal infrastructure*

- Provide a robust, portable ventilator system for facilities that lack sophisticated hospital infrastructure
- A system that requires minimum training and provides ventilation support from initiation to weaning with full range of mainstream parameters
- A system that can supplement the scarce, expensive ICU Ventilators, sparing them for more complex cases
- A system within the budget of the bottom of the pyramid at a fraction of the cost of an ICU Ventilator
- A system to bridge the gap between nothing and an expensive ICU ventilator
- A system that works in harsh uncontrolled environments and as a travel ventilator

INSPIRE-100 Details

System Components

Technical Details

Elegance in Simplicity

Simple, Easy-to-read Front Control Panel

Prominent, Always-on Parameters Display

Menu-driven HMI

Prominent, Tactile Control Buttons

LCD Screen for Menu and Message Display

Colored LEDs show System state at-a-glance

Brightness and Buzzer Volume Controls

Backed by Comprehensive Remote Displays

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Product Overview

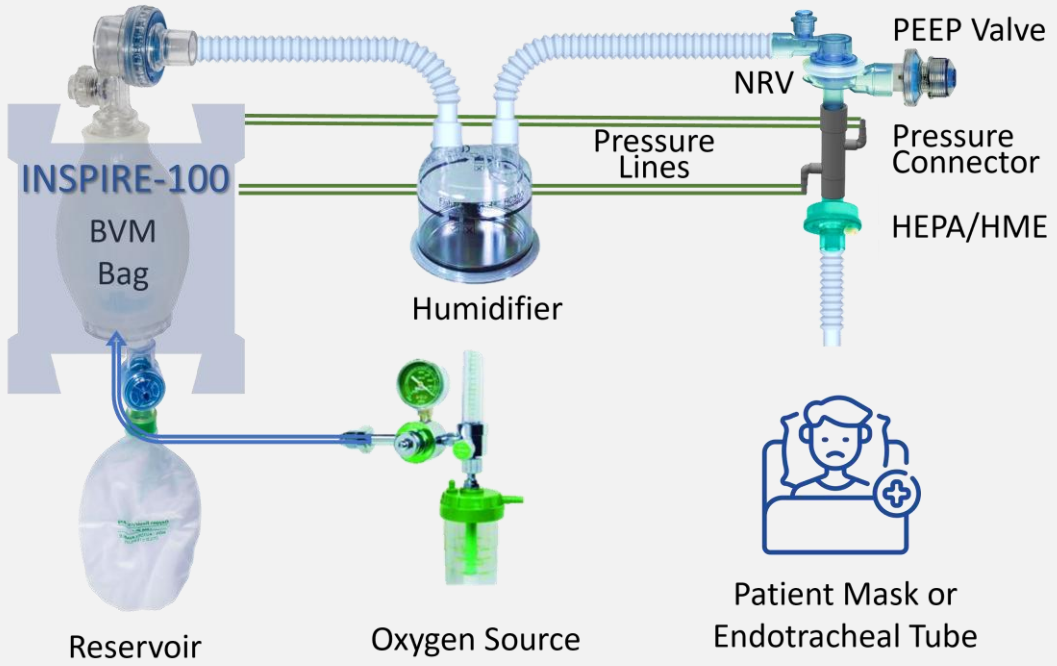
Salient Features

- *Exceptionally Budget Friendly*
- *Comprehensive Remote Monitoring*
- *Easy-to-use Easy-to-train Rugged & Robust*
- *Works without Compressed air or Piped O_2*
- *Complete Set of Mainstream Respiration Parameters*
- *Power Consumption 100W*

Patient Comfort

- *Mandatory Breaths synchronized with Patient-initiated Breaths*
- *No breath stacking*

Breathing Circuit



$$Q \propto \sqrt{\frac{(P_{G1}-P_{G2})}{(P_{G1}+P_{G2})+2*Patmosphere}}$$

An important and necessary simplification is that P_{G1} and P_{G2} encounter[illegible] ur system are of the order of tens of cmH_2O while *Patmosphere* is of the order of a th[illegible] cmH_2O pressure. At sea level, *Patmosphere* is approximately 1000 cmH_2O. Even at an a[illegible]de o[illegible]000 feet, *Patmosphere* is approximately 600 cmH_2O. On the o[illegible] P_{G1} and P_{G2} in the system range from 1 cmH_2O to 60 cmH_2O .

Thus the $(P_{G1} + P_{G2})$ term [illegible] d to $(2* Patmosphere)$, even[illegible] since it is preceded by a square ro[illegible] The [illegible] can be simplified to the one below

$$Q \propto \sqrt{\frac{(P_{G1}-P_{G2})}{[illegible]here}}$$

Re[illegible] *Equation 2* from the theory sectio[illegible] this [illegible]uation can be recast as below given that t[illegible]ce characteristics and pressu[illegible] are the same for every system.

$$Q = C * \frac{\sqrt{(P_{G1}-P[illegible])}}{\sqrt{(Patmos[illegible]ere)}}$$ [illegible] *C = f(Re) Reynold's number*

At a given geographical location, *Patmosphere* is also a constant. So, the above equation further reduces to the one below.

$$Q = \left(\frac{C}{\sqrt{Patmoshpere}}\right) * \sqrt{(P_{G1} - P_{G2})}$$

The equation needs further simplification to ease the computation burden of the square root computation for and inexpensive micro-controller. The constraints are as below.

US Patent Granted

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Technical Specifications

Commonly used Ventilation Modes

CMV	Continuous Mandatory Ventilation
ACV	Synchronized Assist Control Ventilation
SIMV	Synchronized Intermittent Mandatory Ventilation
PSV	Pressure Support Ventilation (BiPAP equivalent)

Full Set of Alarm Alerts

Max Pressure	Pressure Leak	Pressure Loss
Airway Blockage	Coughing Hiccupping	Inconsistent Parameters
Extreme Parameters	System Temperature	And many more …

Volume Controlled Breaths

Tidal Volume **200 - 600 ml**	***Respiratory Rate*** **10 - 30 bpm**	***I:E Ratio*** **1:1 1:2 1:3**
PEEP **4 - 15 cmH_2O**	***Max Pressure*** **15 - 60 cmH_2O**	***FiO_2 Support*** **System Managed** **Externally Controlled**

Pressure Supported Breaths

Support Pressure	5 - 20 cmH_2O
Support Pressure Termination	Flow Triggered 10 – 60%

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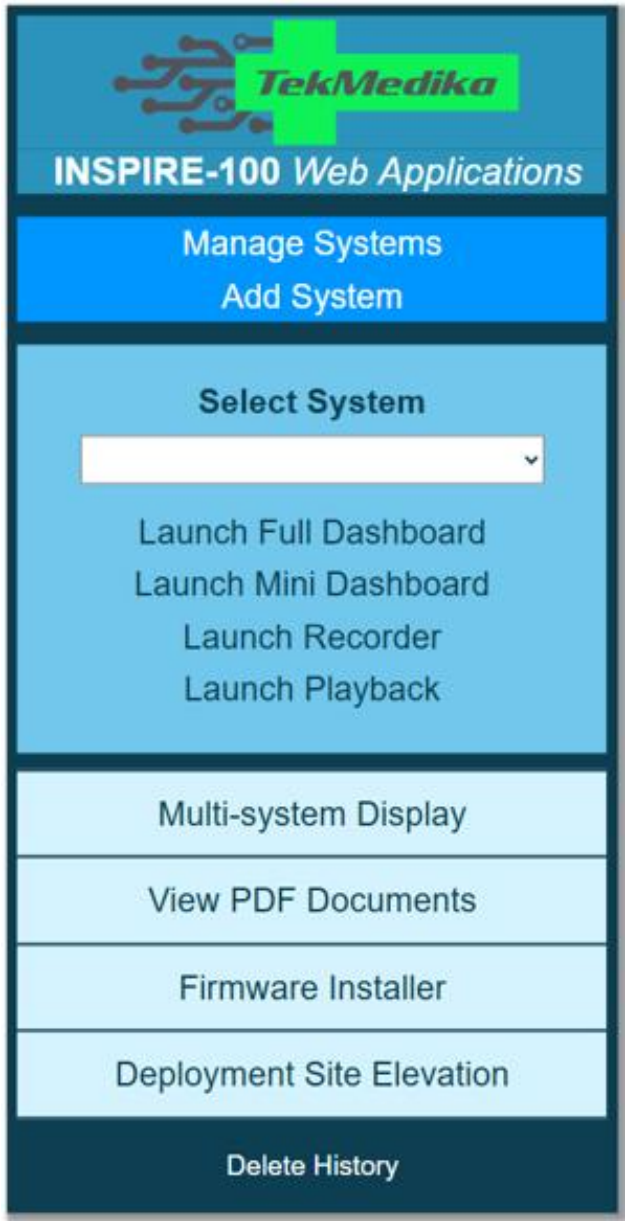
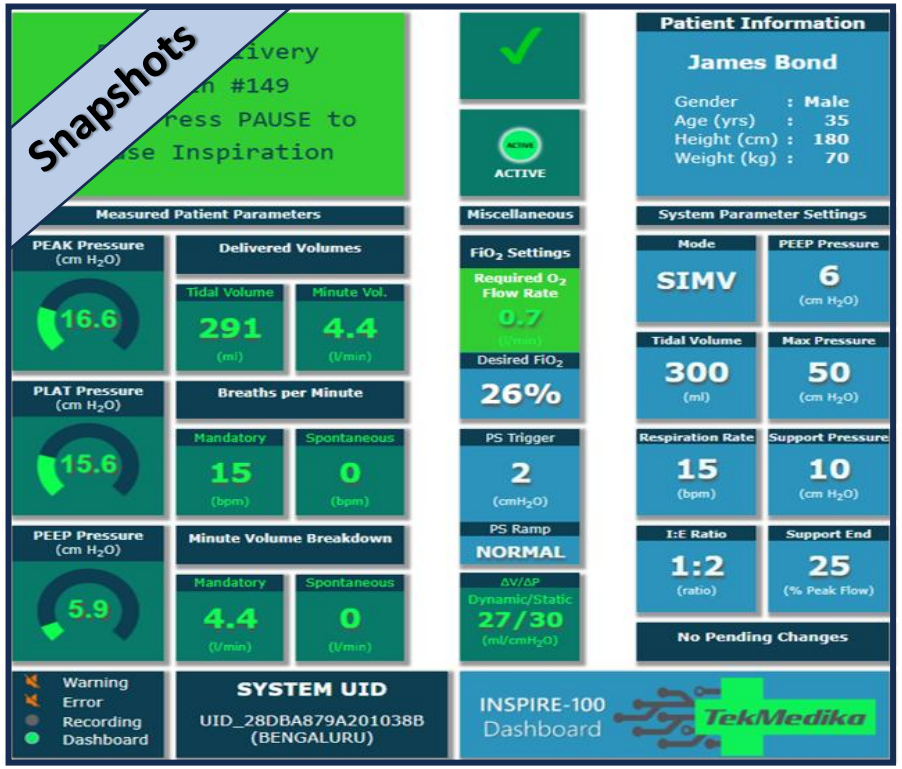
Elaborate Remote Monitoring

- Live Dashboard
- Detailed Breath Waveforms
- Charts for all Parameters
- Detailed Statistics
- System Alerts and Alarms
- Recording and Playback
- Multi-system Display

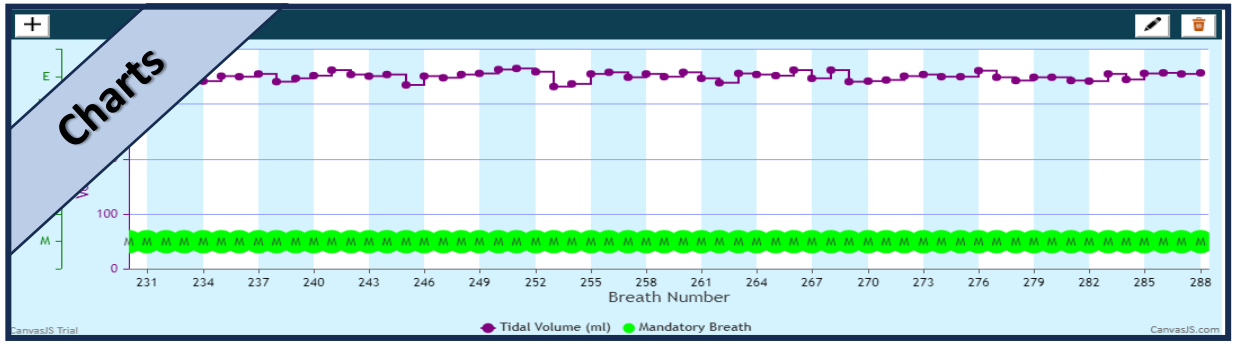


Remote Monitor Screenshots

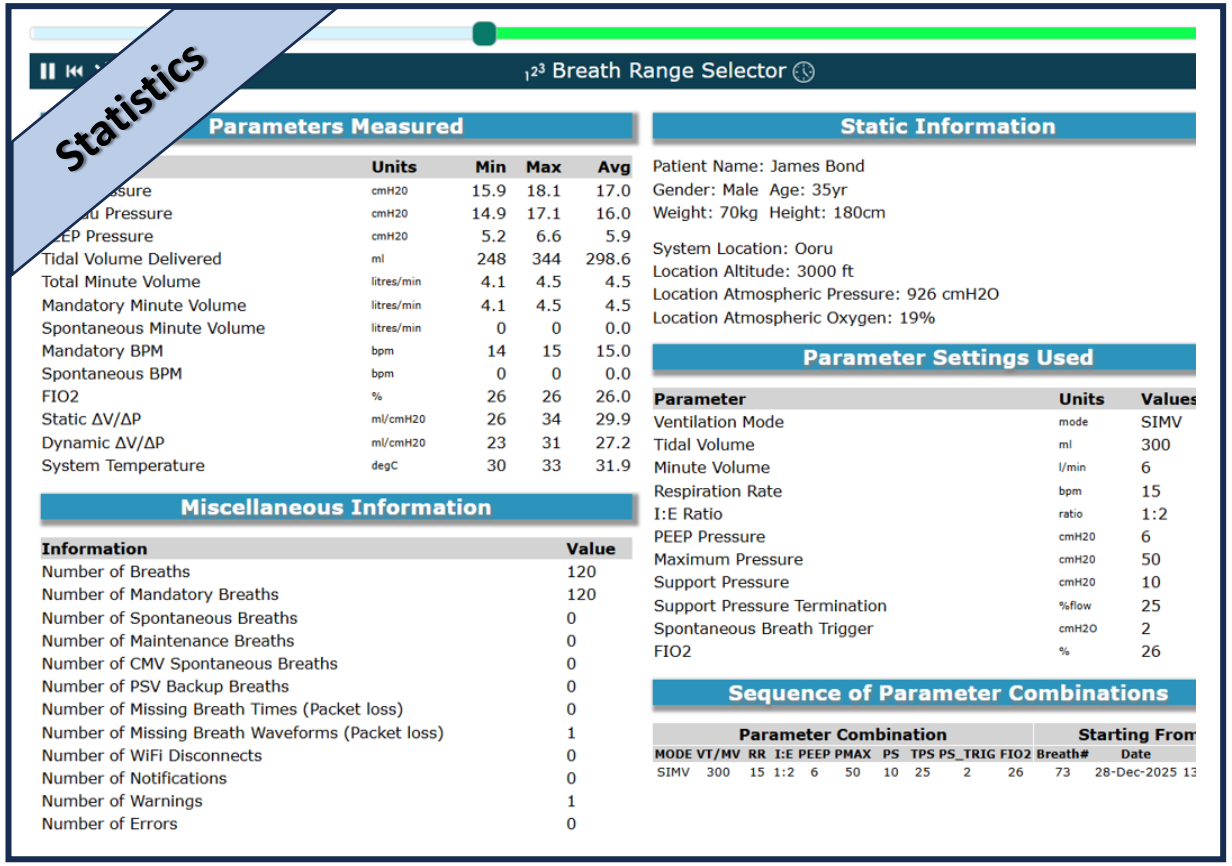
Snapshots



Charts



Statistics



Breath Range Selector

Parameters Measured

	Units	Min	Max	Avg
ssure	cmH20	15.9	18.1	17.0
u Pressure	cmH20	14.9	17.1	16.0
EP Pressure	cmH20	5.2	6.6	5.9
Tidal Volume Delivered	ml	248	344	298.6
Total Minute Volume	litres/min	4.1	4.5	4.5
Mandatory Minute Volume	litres/min	4.1	4.5	4.5
Spontaneous Minute Volume	litres/min	0	0	0.0
Mandatory BPM	bpm	14	15	15.0
Spontaneous BPM	bpm	0	0	0.0
FIO2	%	26	26	26.0
Static ΔV/ΔP	ml/cmH20	26	34	29.9
Dynamic ΔV/ΔP	ml/cmH20	23	31	27.2
System Temperature	degC	30	33	31.9

Miscellaneous Information

Information	Value
Number of Breaths	120
Number of Mandatory Breaths	120
Number of Spontaneous Breaths	0
Number of Maintenance Breaths	0
Number of CMV Spontaneous Breaths	0
Number of PSV Backup Breaths	0
Number of Missing Breath Times (Packet loss)	0
Number of Missing Breath Waveforms (Packet loss)	1
Number of WiFi Disconnects	0
Number of Notifications	0
Number of Warnings	1
Number of Errors	0

Static Information

Patient Name: James Bond
Gender: Male Age: 35yr
Weight: 70kg Height: 180cm

System Location: Ooru
Location Altitude: 3000 ft
Location Atmospheric Pressure: 926 cmH2O
Location Atmospheric Oxygen: 19%

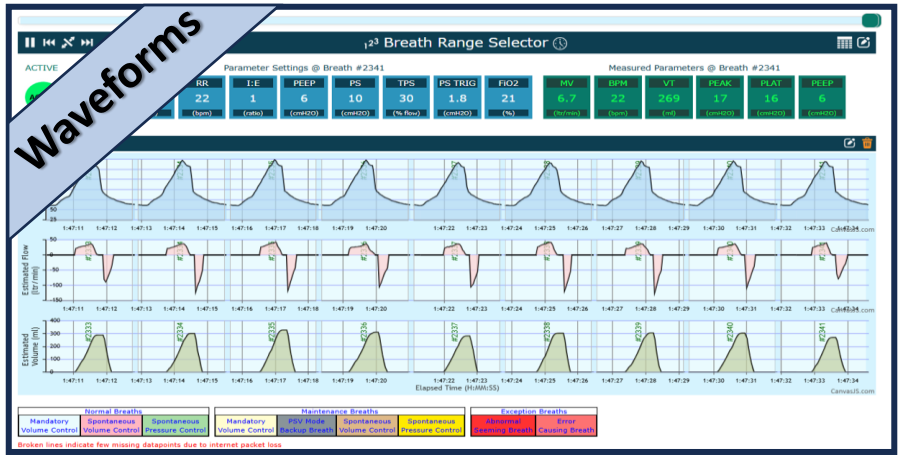
Parameter Settings Used

Parameter	Units	Values
Ventilation Mode	mode	SIMV
Tidal Volume	ml	300
Minute Volume	l/min	6
Respiration Rate	bpm	15
I:E Ratio	ratio	1:2
PEEP Pressure	cmH20	6
Maximum Pressure	cmH20	50
Support Pressure	cmH20	10
Support Pressure Termination	%flow	25
Spontaneous Breath Trigger	cmH2O	2
FIO2	%	26

Sequence of Parameter Combinations

Parameter Combination											Starting From	
MODE	VT/MV	RR	I:E	PEEP	PMAX	PS	TPS	PS_TRIG	FIO2	Breath#	Date	
SIMV	300	15	1:2	6	50	10	25	2	26	73	28-Dec-2025	13

Waveforms

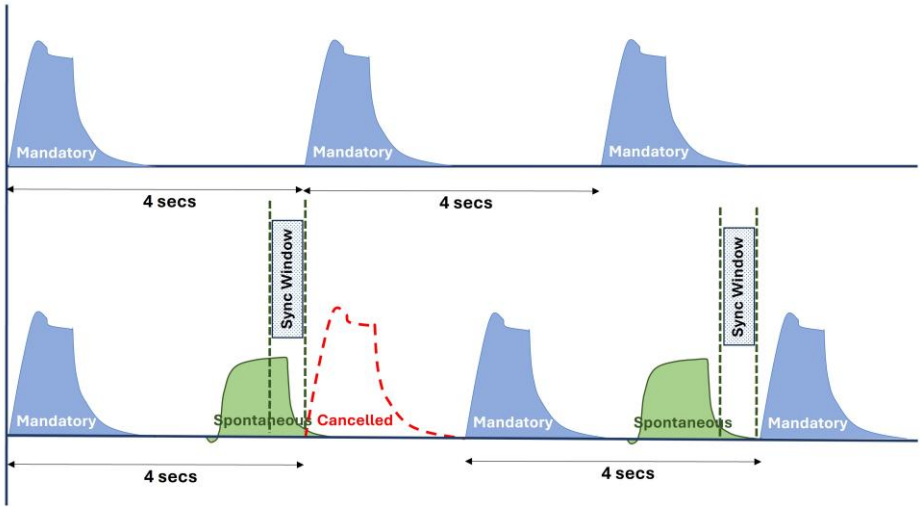


Breath Synchronization

Synchronize Mandatory breaths with Spontaneous breaths
Prevent breath stacking

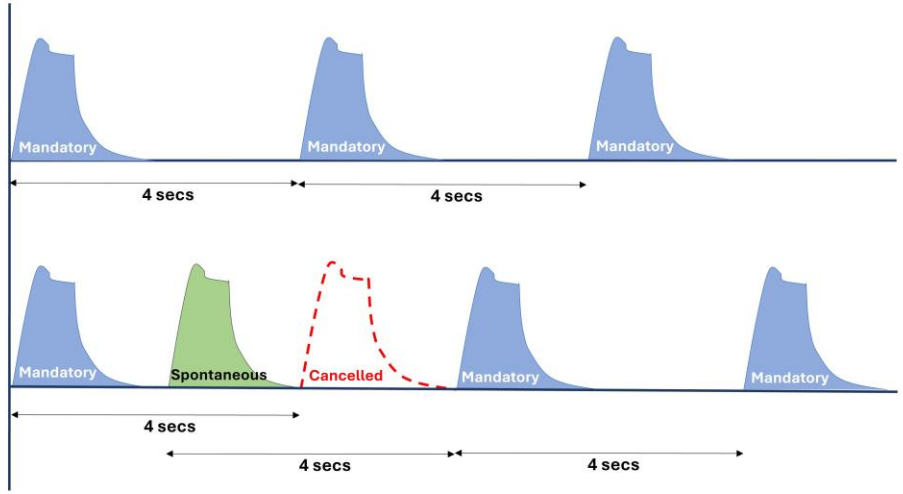
Breath Synchronization in SIMV Mode

e.g. RR=15 bpm (4 secs per breath)



Breath Synchronization in ACV Mode

e.g. RR=15 bpm (4 secs per breath)



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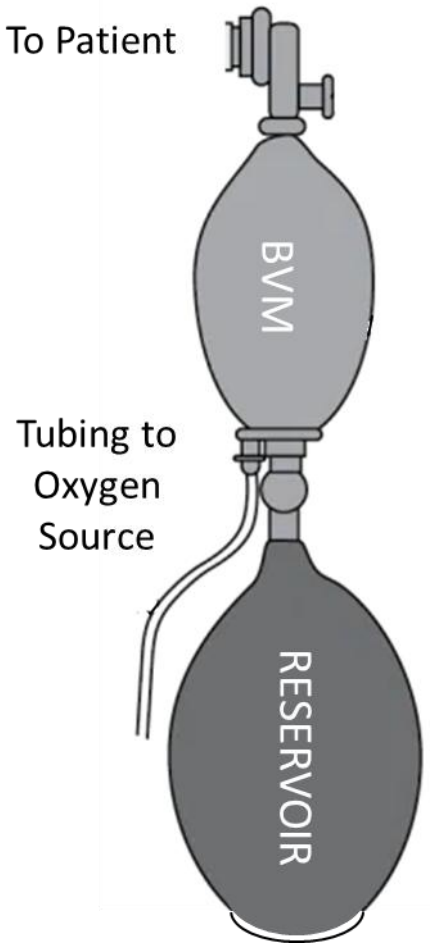
FiO_2 Settings

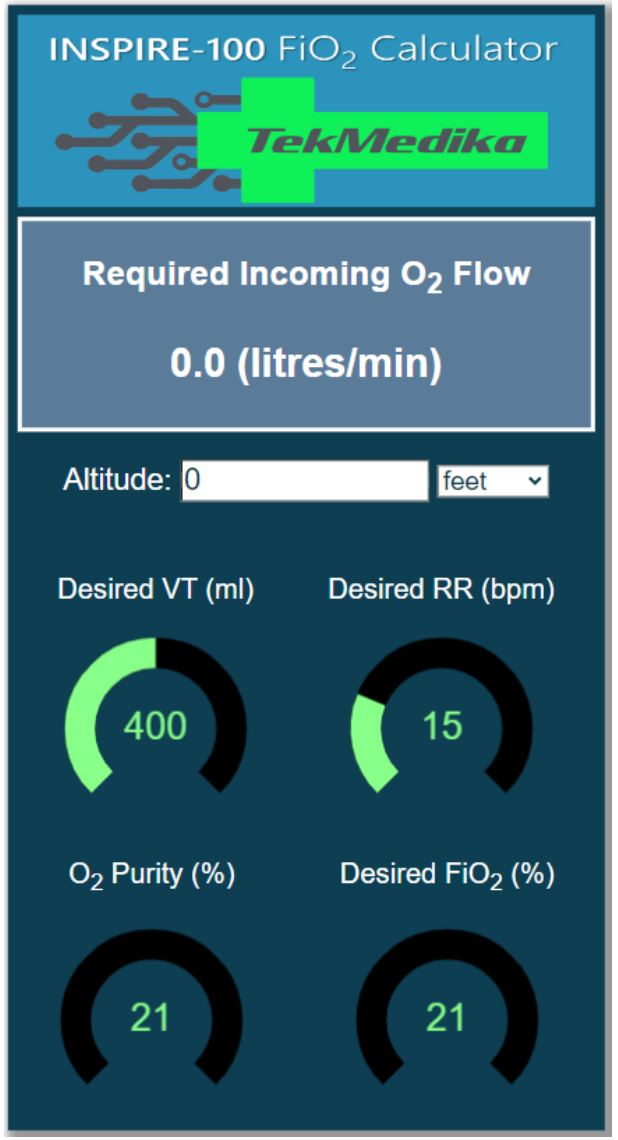
Without the Reservoir bag, FiO_2 delivered is the Atmospheric O_2 content at site

FiO_2 delivery with the Reservoir bag is mathematically modelled, calibrated and verified in the Lab to provide +/- 10% accuracy

Front-panel guides the user in setting the appropriate input O_2 flow rate from the O_2 source for a given FiO_2

The mathematical model provides for an O_2 concentrator as an O_2 source (purity < 100%)





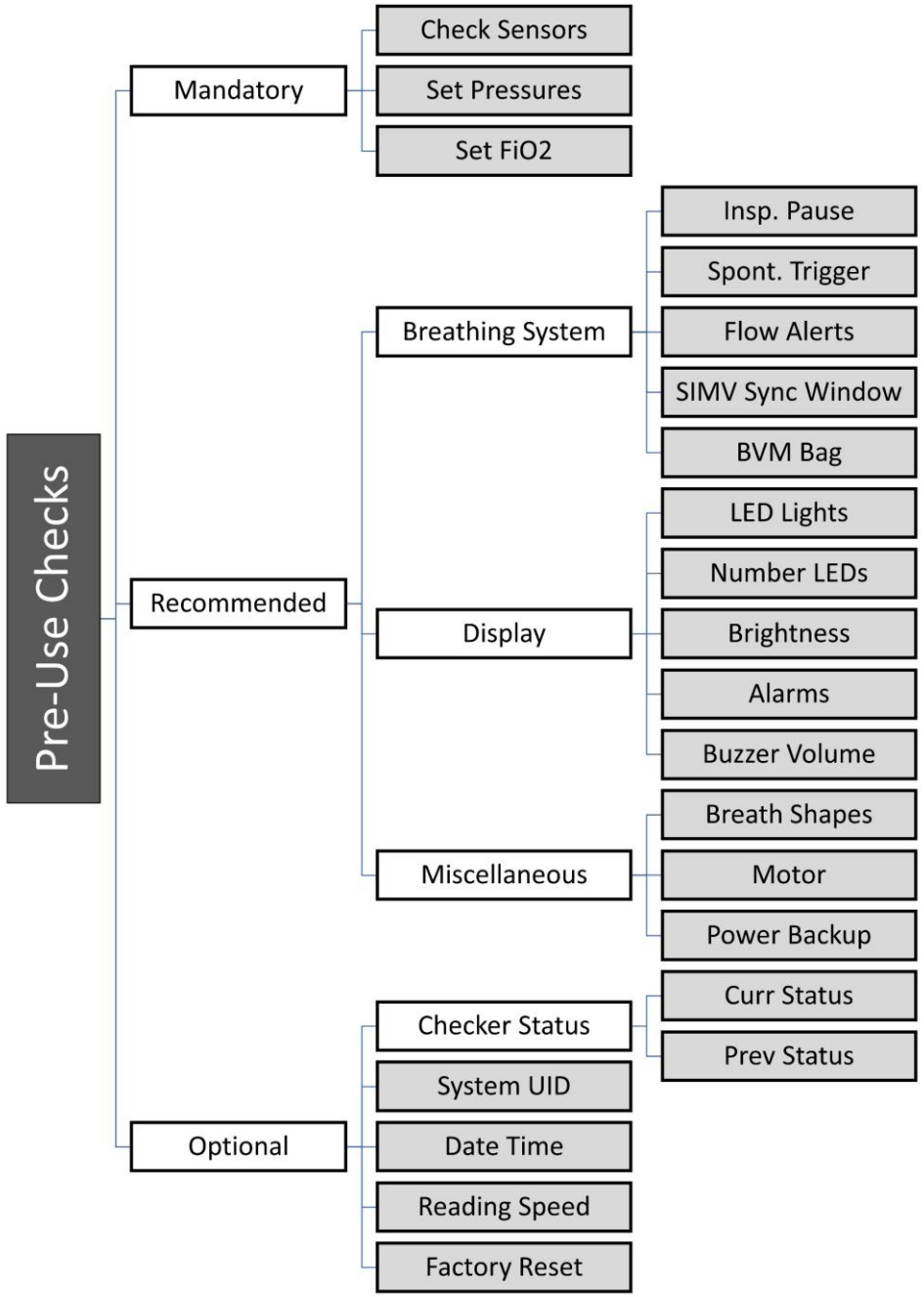
Alarms & Safety Features

Enforcement of Pre-use checks

Maintenance Breaths till Alarm situation rectified

Alarms, Warnings and Notifications

- Max Pressure Alarm
- Pressure Leak Alarm
- Pressure Loss Alarm
- Airway Blockage Alarm
- System Temperature Alarm
- Sensor Failure Alarm
- Breathing Circuit Failure Alarm
- Detect coughing/hiccupping fits
- Inconsistent input parameters
- Extreme parameter combination warnings
- And many more …



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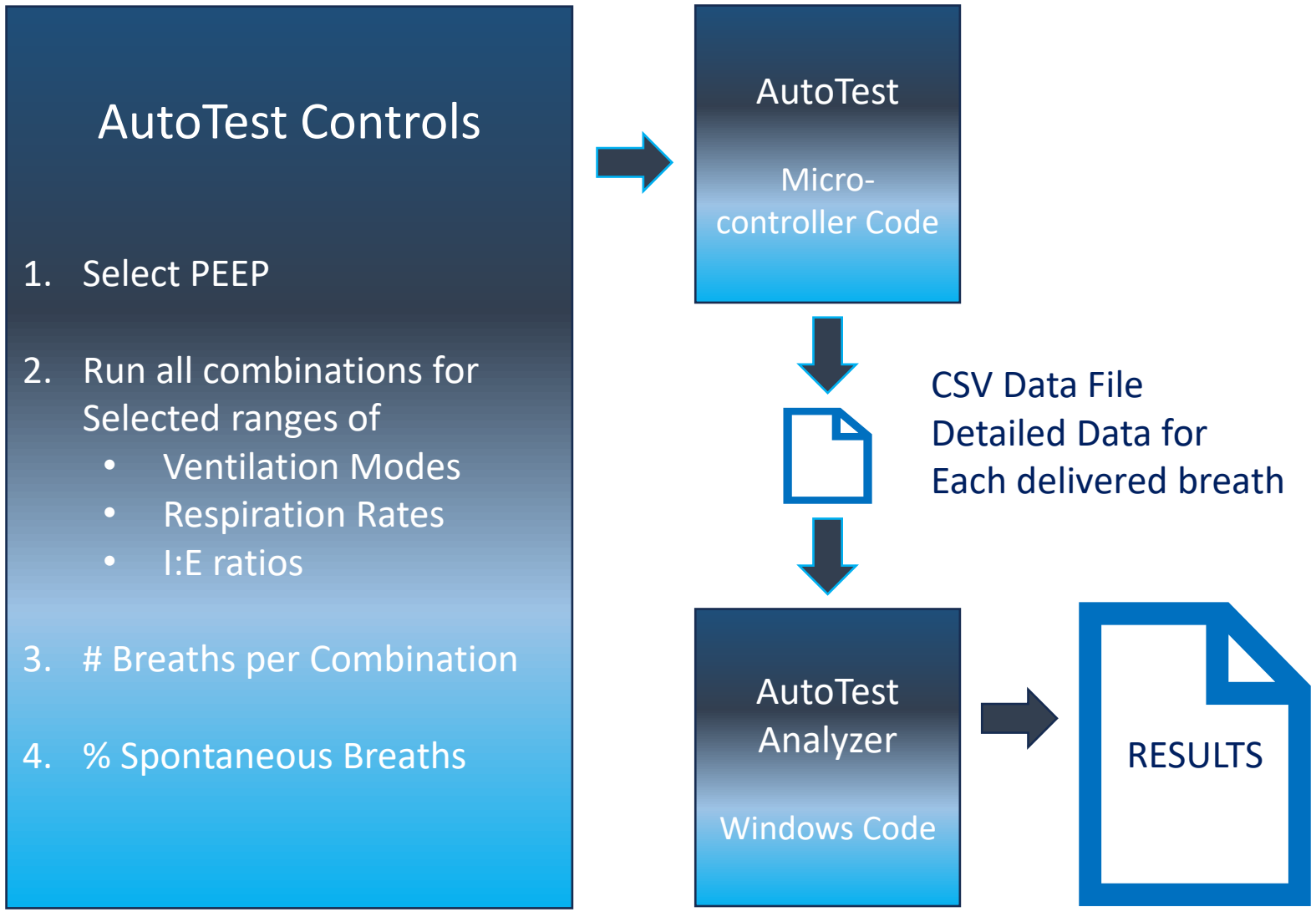
Extensive Automated Testing

AutoTest Controls

1. Select PEEP
2. Run all combinations for Selected ranges of
   - Ventilation Modes
   - Respiration Rates
   - I:E ratios
3. # Breaths per Combination
4. % Spontaneous Breaths



BACKUP

CPAP *v/s* BiPAP *v/s* INSPIRE-100 *v/s* ICU-VENTILATOR

CPAP	BiPAP	Inspire 100	ICU Ventilator
Continuous Positive Airway Pressure	Continuous Bi-Level Airway Positive Pressure	Mechanical Ventilation with 4 most-used ventilation modes and controls	Mechanical Ventilation with very sophisticated modes and controls
Non-invasive	Non-invasive	Non-invasive + Invasive	Non-invasive + Invasive
High Flow + PEEP	Inspiratory Pressure + PEEP	Tidal Volume + Support Pressure + PEEP	Tidal Volume + Support Pressure + PEEP
Useful for Type 1 respiratory Failure (Hypoxemic)	Useful for Type 2 respiratory Failure (Hypercopnic)	Useful for Hypoxemic and Hypercopnic respiratory failure	Useful for Hypoxemic and Hypercopnic respiratory failure
Continuous flow of air at a constant pressure. Increases mean airway pressure to recruit collapsed alveoli	Continuous flow of air at different constant pressures during inspiration and expiration breathing phase	Independent control over the volume, the respiration rate and pressure	Independent control over the volume, the respiration rate and pressure
Useful only when patient can breathe on his own	Useful only when patient can breathe on his own	Useful when patient can or CANNOT breathe on his own	Useful when patient can or CANNOT breathe on his own
Only Spontaneous breaths that are patient triggered.	Only Spontaneous breaths that are patient triggered.	Spontaneous breaths + Mandatory breaths controlled by RR and I:E	Spontaneous breaths + Mandatory breaths controlled by RR and I:E
External FiO2 control	External FiO2 control	System assisted FiO2 control	Direct FiO2 control
Breath Synchronization N/A	Breath Synchronization N/A	Full Breath Synchronization	Full Breath Synchronization
No Tidal Volume control	Indirect Tidal Volume control (IPAP-EPAP)	Direct Tidal Volume control	Direct Tidal Volume control
No Respiration Rate control	No Respiration Rate control	Direct Respiration Rate control	Direct Respiration Rate control
No Inspiration:Expiration ratio control	No Inspiration:Expiration ratio control	Direct Inspiration:Expiration control	Direct Inspiration:Expiration control
External Humidity control	External Humidity control	External Humidity control	Direct Humidity control
No display of Peak, Plateau or PEEP	No display of Peak, Plateau or PEEP	Full display of Peak, Plateau and PEEP	Full display of Peak, Plateau and PEEP
Minimal alarm signals	Minimal alarm signals	Full set of Alarm signals	Full set of Alarm signals
No remote monitoring	No remote monitoring	Sophisticated Remote WEB Dashboard	Minimal Remote monitoring (if any)

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Thank You